This raw curve can be converted into the complete Gaussian by adding an adjustable mean, μ, and standard deviation, σ. In addition, the equation must be normalized so that the total area under the curve is equal to *one*, a requirement of all probability distribution functions. This results in the general form of the normal distribution, one of the most important relations in statistics and probability:

EQUATION 2-8
Equation for the *normal distribution*, also called the *Gauss distribution*, or simply a *Gaussian*. In this relation, $P(x)$ is the probability distribution function, μ is the mean, and s is the standard deviation.

$$P(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-(x-\mu)^2/2\sigma^2}$$

Figure 2-8 shows several examples of Gaussian curves with various means and standard deviations. The *mean* centers the curve over a particular value, while the *standard deviation* controls the width of the bell shape.

An interesting characteristic of the Gaussian is that the *tails* drop toward zero very rapidly, much faster than with other common functions such as decaying exponentials or $1/x$. For example, at two, four, and six standard

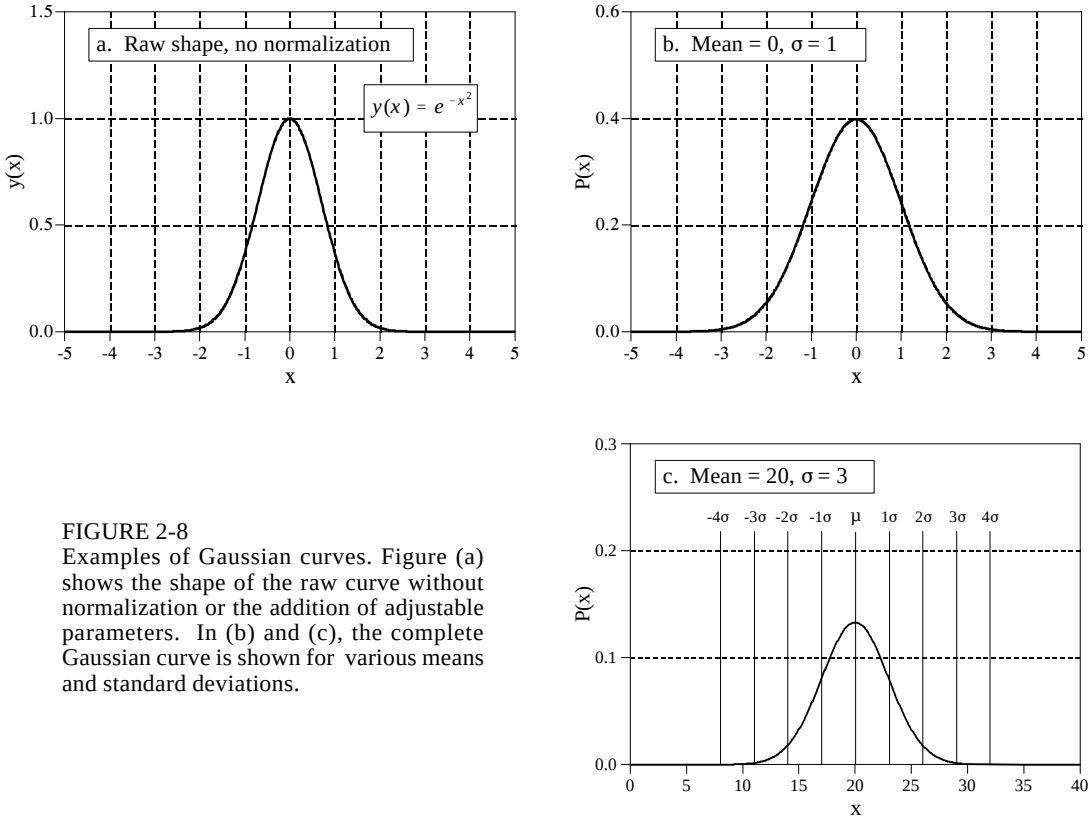

FIGURE 2-8
Examples of Gaussian curves. Figure (a) shows the shape of the raw curve without normalization or the addition of adjustable parameters. In (b) and (c), the complete Gaussian curve is shown for various means and standard deviations.